INCH-POUND
MIL-DTL-24787A(SH)
24 September 2020
SUPERSEDING
MIL-F-24787(SH)
3 September 1993

DETAIL SPECIFICATION

FITTINGS, END, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers reusable end fittings for use with flexible rubber hose assemblies in low, medium, and high pressure shipboard piping systems.

1.2 Classification.

1.2.1 Types. The following end fitting types are covered by this specification as specified (see 6.2):

End fitting type	Description	Specification sheet
A	37-degree flare swivel to hose, straight.	/2
C	O-ring seal union, male half to hose, straight.	/3
CL	O-ring seal union, male half to hose, 90-degree elbow.	/3
D	Hose to hose, 90-degree elbow.	/5
E	Split clamp to hose, straight.	/4
EL	Split clamp to hose, 90-degree elbow.	/4
F	Flange to hose, straight.	/1
FL	Flange to hose, 90-degree elbow.	/1
FFL	Flange to hose, 45-degree elbow.	/1
U	Hose to hose,180-degree return.	/6
SC	Split clamp assembly.	/4
TF	Tailpiece, female, split clamp, silver braze.	/4
TM	Tailpiece, male, split clamp, silver braze.	/4

1.2.2 Hose specification grouping. Hose specification groups (see 6.2) identify the particular hose (MIL-DTL-24135 and MIL-DTL-24136) for which a specific end fitting design has been produced. See 6.4 for hose commonality data.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4730

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

(See ASSIST database for list of specification sheets.)

MIL-PRF-2104	-	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-H-19457	-	Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic
MIL-DTL-24135	-	Hose, Synthetic Rubber, Wire Reinforced, for Flexible Hose Assemblies, General Specification for
MIL-DTL-24135/12	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Single-Wire Braid Over Single Synthetic Fiber Braid), for Flexible Hose Assemblies (Medium Pressure, High Temperature, Multi-Fluid, Sizes -4 through -40)
MIL-DTL-24135/13	-	Hose, Synthetic Rubber, High Performance, Wire Reinforced (Double-Wire Braid), for Flexible Hose Assemblies (High Pressure, High Temperature, Multi-Based Oil, Sizes -4 through -32)
MIL-DTL-24136	-	Hose, Synthetic Rubber, Synthetic Fiber Reinforced, for Flexible Hose Assemblies, General Specification for
MIL-DTL-32353	-	Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-276 - Impregnation of Porous Metal Castings and Powdered Metal Components

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online from the Technical Data Management Information System (TDMIS) or online at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D380 - Standard Test Methods for Rubber Hose

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the applicable specification sheet, the latter shall govern.

3.2 Qualification. End fittings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 Material.

3.3.1 End fitting materials. End fittings shall be of a material as specified on the applicable specification sheet.

3.3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and construction.

3.4.1 End fitting characteristics. End fittings shall be of the types and sizes specified on the figures and in the tables of the applicable specification sheet and shall be of a reusable design. Hose end of the fitting may be of a solid socket, ring-retained segmented socket, bolt retained segmented socket, or any other design capable of meeting all the requirements of this specification and the applicable specification sheet. Hose end of the fitting shall be readily removable and re-attachable to a new section of hose by the use of commonly available tools that are normally carried aboard ship. The need for special tools shall be subject to the review and approval of NAVSEA.

3.4.2 Dimensions. Dimensions of the end fittings shall not exceed the envelope limits shown on the figures of the applicable specification sheet.

3.4.3 Finish. End fittings shall be finished smooth and free from nicks, scratches, burrs, and sharp edges.

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3.4.4 Tool and die marks (nickel copper alloy barstock only). The presence of tool marks on the machined parts is acceptable provided such marks do not exceed 3 percent of the wall thickness. Draw marks upon the external hexagonal surfaces, such as the outside diameter (OD) of solid sockets, which are normally un-machined surfaces, are acceptable if they do not exceed 3 percent of the wall thickness.

3.4.5 Welding. Welding, if used, shall be in accordance with S9074-AR-GIB-010/278. Brazing of end fitting parts shall not be permitted. If welding is used in the fabrication of any of the types of end fittings covered by this specification, welding procedures and classifications in accordance with S9074-AR-GIB-010/278 used by the contractor for the materials specified herein shall be approved by NAVSEA prior to the start of qualification testing.

3.4.5.1 Weld repair. Weld repair shall be in accordance with S9074-AR-GIB-010/278.

3.4.6 Impregnation repair. Impregnation repair of porous metal castings shall be in accordance with MIL-STD-276.

3.4.7 Flanges. Flanges are considered to be non-pressure containing and are not required to be hydrostatically tested as an individual piece.

3.4.8 Mating of parts. Where two parts of the same material are mated together (e.g., fasteners), provisions shall be made to prevent galling and seizing of the mated surfaces. A solid film lubricant, such as molybdenum disulfide, or two classes of heat treatments of the same material, or a combination of any of these methods shall be used to prevent galling and seizing. Parts that will come into direct contact with the inner bore of the hose are not considered to be mating parts (i.e., nipple).

3.5 Physical requirements.

3.5.1 Proof pressure. End fittings shall not separate from the hose nor show evidence of leakage or deformation when subjected to the proof pressure test specified in 4.5.1.

3.5.2 Oil circulation. The end fittings shall show no signs of leakage or separation from the hose when subjected to the oil circulation test specified in 4.5.2.

3.5.3 Impulse. The end fittings shall show no signs of leakage or separation from the hose when subjected to the impulse test specified in 4.5.3.

3.5.4 Burst test. End fittings shall show no signs of leakage or separation from the hose at pressures less than the specified burst pressure of the hose to which the end fitting is attached when tested as specified in 4.5.4.

3.6 Marking. Each end fitting shall have permanent and visible stamped identification. The numerals and lettering shall be as large as practical, but not less than 1/16 inch high and may be placed on more than one flat. The identification marking shall contain the following:

a. Specification sheet number (e.g., “24787/4”).

b. Manufacturer’s part number.

c. Pipe or tube size × hose dash size (e.g., “2½ × -40”).

d. Manufacturer’s CAGE code.

e. Manufacturer’s trademark or logo (optional).

f. End fitting type and hose specifications.

g. Flanges shall have the flange class on the OD surface.

3.7 Installation instructions. Instructions for installing the end fitting on to the hose should be as specified (see 6.2).

3.8 Threads. Where screw threads are required, including bolting for bolt together segmented end fittings, they shall be in accordance with FED-STD-H28.

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3.9 Workmanship. End fittings shall be uniform in consistency and in material. Castings shall be free from patching, misalignment resulting from shifted coring, warping, porosity, or other defects. Sprues shall be removed, and the castings shall be free from sand, dirt, scale, and other extraneous materials. Machined parts shall be manufactured to tolerances and dimensions specified herein and on the applicable specification sheet. End fittings shall be free from workmanship deficiencies that could impair the function or serviceability of the hose and end fitting in its intended use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examination of 4.4 and the tests of 4.5.

4.2.1 Sampling for qualification. Sampling of end fittings for assembly of test hose assemblies and the sequence of tests shall be as specified in the applicable specification sheet.

4.2.2 End fitting types. If the end fitting hose attachment (e.g., solid socket, ring and band) is identical for more than one type of end fitting, only a representative type of end fitting hose attachment will be required for qualification testing. Qualification may be granted to all those types that the sample represents, subject to results of the tests.

4.3 Conformance inspection. Conformance sampling, inspection, and testing shall consist of the following:

a. End fittings (see 4.3.1).

b. Hose assembly(ies) (see 4.3.2).

4.3.1 End fittings. Conformance inspection of end fittings shall be in accordance with the visual and dimensional examination of 4.4. No tests are required of end fittings for lot acceptance testing.

4.3.1.1 Lots. For the purpose of conformance inspection, a lot is defined as all of the end fittings of the same size and type, produced in one facility, using the same production processes and materials, and being offered for delivery at one time.

4.3.1.2 Sampling for end fitting examination. As a minimum, a sample quantity from each lot of completed end fittings shall be selected in accordance with table I.

TABLE I. Sampling for visual and dimensional examination.

Lot size	Sample size
2 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1,200	15
1,201 to 3,200	18
3,201 to 10,000	22

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4.3.2 Hose assembly(ies). Conformance inspection of hose assemblies shall include the sample quantity of 4.3.2.1, the visual and dimension examination of 4.4, and the proof test of 4.5.1. Visual and dimensional examination of the end fittings shall occur prior to constructing the hose assembly.

4.3.2.1 Sampling for tests. Visual and dimensional examination and proof testing is required of all hose assemblies in the purchase order, regardless of type or size.

4.4 Examination. Each sample end fitting selected in accordance with 4.2.1 or 4.3.2 shall be visually and dimensionally examined to determine conformance with the requirement paragraphs listed in table II.

TABLE II. Visual and dimensional examination of sample end fitting.

Examination	Requirement
Material	3.3
Design and construction	3.4
Marking	3.6
Installation instructions (see 6.2)	3.7
Threads	3.8
Workmanship	3.9

4.4.1 Lot rejection. Any sample having one or more visual or dimensional defect shall be rejected and shall be cause for rejection of the entire lot represented by the sample. Rejection of a lot shall require that corrective action be implemented by the contractor.

4.5 Tests.

4.5.1 Proof pressure. The proof pressure test of the hose assemblies containing the sample end fittings shall be conducted in accordance with ASTM D380. Samples may be pneumatically or hydrostatically tested. Pneumatic testing must be conducted with the hose assembly submerged in water. Hydrostatic testing may be conducted with water, oil, or hydraulic fluid in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, MIL-PRF-83282, or MIL-H-19457 (applicable to MIL-DTL-24135/12 and MIL-DTL-24135/13 hoses only). The proof pressure shall be 200 percent of the maximum working pressure specified in the applicable specification sheet and shall be held for not less than 1 minute nor greater than 5 minutes. The pressure shall be raised at a uniform rate as follows:

a. For sizes -32 and smaller, the rate of pressure increase shall be not less than 15,000 pounds per square inch (psi) nor greater than 25,000 psi per minute.

b. For sizes greater than -32, the pressure increase shall not exceed 1,000 psi per minute.

Evidence of leakage past the end fitting or separation of the end fitting from the hose shall constitute failure of this test.

4.5.2 Oil circulation. Sample hose assemblies shall be filled with oil in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, MIL-PRF-83282, or MIL-H-19457 (applicable to MIL-DTL-24135/12 and MIL-DTL-24135/13 hoses only) and pressurized to between 75 and 100 psi. Unless otherwise specified in the applicable specification sheet, the temperature shall be raised from ambient to 180±10 °F and the fluid circulated through the hose assembly at a flow rate of not less than 3 gallons per minute. The ambient temperature shall be maintained at 75±15 °F. Every 24 hours, the pressure shall be increased to the maximum working pressure listed in the applicable specification sheet and maintained for not less than 5 minutes. The flow shall be continued for not less than 200 hours. At the completion of this test, the hose assembly shall be pressurized to the maximum working pressure specified in the applicable specification sheet with the pressure held for not less than 5 minutes. Nonconformance to 3.5.2 shall constitute failure of this test.

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4.5.3 Impulse. Hose assemblies fitted with end fittings being qualified shall be subjected to impulse testing to the peak pressure specified on the applicable specification sheet. Hose assemblies shall be subjected to an impulse rate of 30 to 80 cycles per minute, with the oil temperature of not less than 120 °F nor greater than 130 °F (unless otherwise specified on the applicable specification sheet) for a minimum of 150,000 cycles. At the beginning of each pressure cycle, the peak pressure specified on the applicable specification sheet shall be attained. The applicable working pressure shall be attained before leveling off (see figures 1, 2, or 3 as applicable).

4.5.3.1 Retest and impulse averaging. In the event of a hose failure during impulse testing, an additional hose assembly shall be tested, and the results of both hose assemblies shall be averaged. The conditions for testing an additional hose assembly are as follows:

a. If the averaging is for a hose qualification as well as an end fitting qualification, the conditions of MIL-DTL-24135 or MIL-DTL-24136 shall apply as applicable.

b. If the test is for end fitting qualification only, the end fitting shall be removed from the failed hose and reset on a new length of hose and the test shall be continued until the required number of cycles has been reached.

If the failure involved end fitting separation from the hose, averaging shall not be permitted. The components of the failed hose assembly shall be analyzed to determine the cause of the end fitting separation from the hose. After corrective action has been taken, a new test using new end fittings and a new hose shall be conducted. Nonconformance to 3.5.3 shall constitute failure of this test.

4.5.4 Burst pressure. The burst pressure test of the hose assembly shall be conducted in accordance with ASTM D380. The pressure shall be raised at a uniform rate as follows:

a. For sizes -32 and smaller, the rate of pressure rise shall be not less than 15,000 psi and not greater than 25,000 psi per minute.

b. For sizes greater than -32, the rate of pressure rise shall not exceed 1,000 psi per minute.

The burst pressure shall be 400 percent of the maximum working pressure specified in the applicable specification sheet. Evidence of leakage, deformation of the end fitting, or end fitting separation at or below the specified test pressure shall constitute failure of this test. End fittings used for the burst test shall not be acceptable for further use. Nonconformance to 3.5.4 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. End fittings are intended for use with hoses in accordance with MIL-DTL-24135 and MIL-DTL-24136 in piping systems as flexible connections to hard mounted and resiliently mounted equipment subject to independent movement. These hoses may be used for other applications that do not exceed the pressure or temperature limits of the hose selected. End fittings procured to this specification used in hose assemblies manufactured with MIL-DTL-24135 and MIL-DTL-24136 hose have an established service life as documented in S6430-AE-TED-010, Piping Devices, Flexible Hose Assemblies. End fittings procured to this specification are form, fit, and function replacements for prior revisions. Existing stock of hose and hose assemblies manufactured to MIL-F-24787, dated 3 September 1993, are acceptable for use until depleted.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and applicable specification sheet.

b. End fitting type (see 1.2.1).

c. End fitting size (pipe end and hose end).

d. When installation instructions are required (see 3.7).

e. Packaging requirements (see 5.1).

f. Hose or hose specification grouping, as applicable (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24787 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Hose commonality data. Hose groupings are intended for specific MIL-DTL-24135 and MIL-DTL-24136 hose specification sheets with common hose sizes. The commonality is shown in table III. Specification sheets that are not listed do not have commonality across the hose specification sheets.

TABLE III. Hose specification sheet commonality.

Common hose dash sizes per specifications and specification sheets
MIL-DTL-24135/1, /5, and /13
MIL-DTL-24135/4 and /10 (-40 and -48 only)
MIL-DTL-24135/10 (-4 through -32), /12, and MIL-DTL-24136/3

6.5 Hose applicability. Where hoses are required for purposes of qualification of end fittings under this specification or for contract or purchase order for hose assemblies, they should be hoses qualified to MIL-DTL-24135 or MIL-DTL-24136.

6.6 Installation instructions. Installation instructions packaged with each end fitting should include the following:

a. List of tools necessary for installation of the end fitting.

b. Sequence of steps for proper attachment of end fitting to hose.

c. Assembly precautions required.

The assembly instructions should be printed on reasonably heavy stock so as to prevent damage during packing and shipping. A copy of the instructions should be furnished to NAVSEA for review as part of the qualification report.

6.7 Subject term (key word) listing.

37-degree flare swivel

90-degree elbow

180-degree return

Flange

Gasket seal

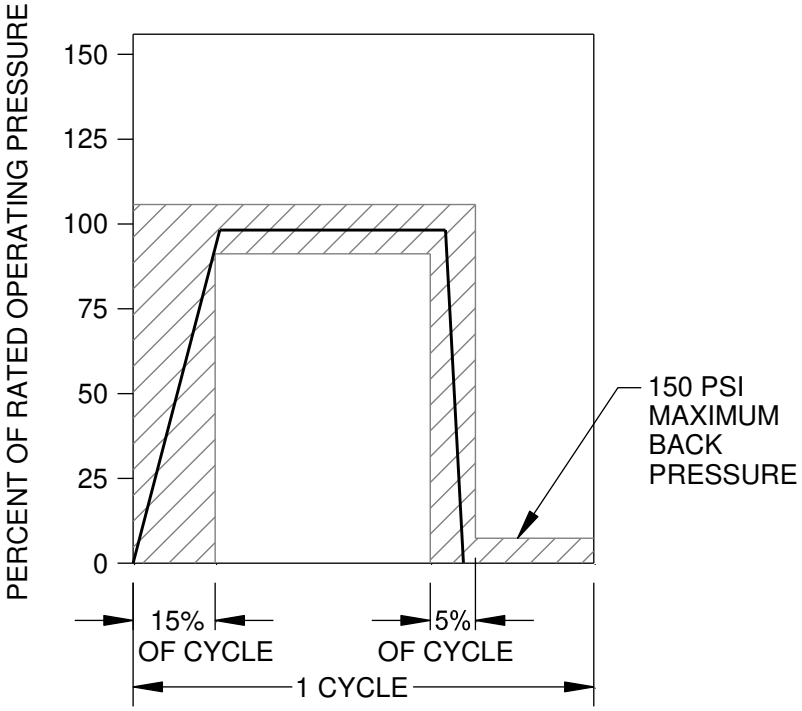
O-ring union

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Split clamp

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

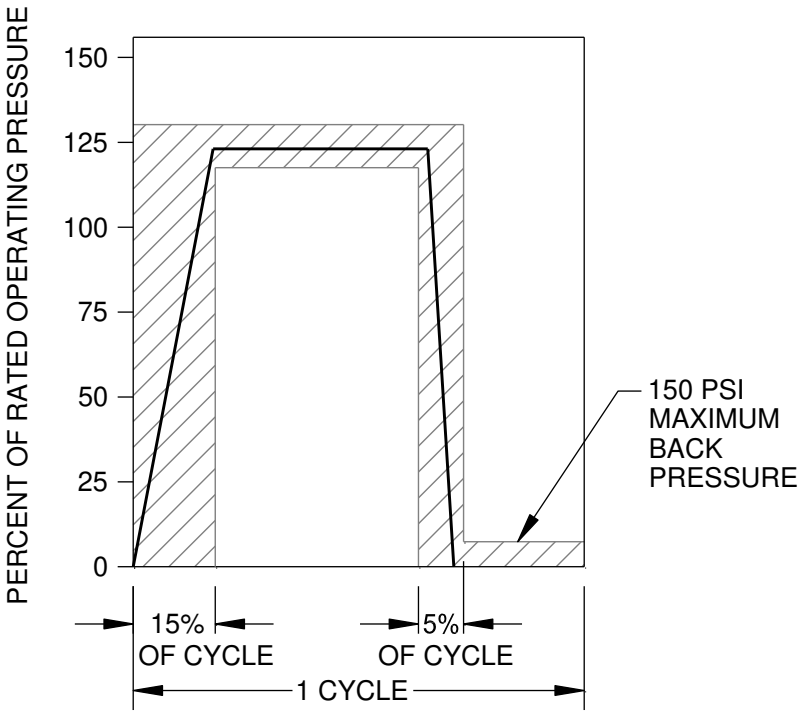
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NOTE: This curve is the approximate pressure time cycle for the pressure impulse to be used in performing the impulse test specified in 4.5.3. The actual pressure time cycle obtained when performing this test should fall within the shaded area.

FIGURE 1. Pressure time curve for impulse test (100 percent square wave).

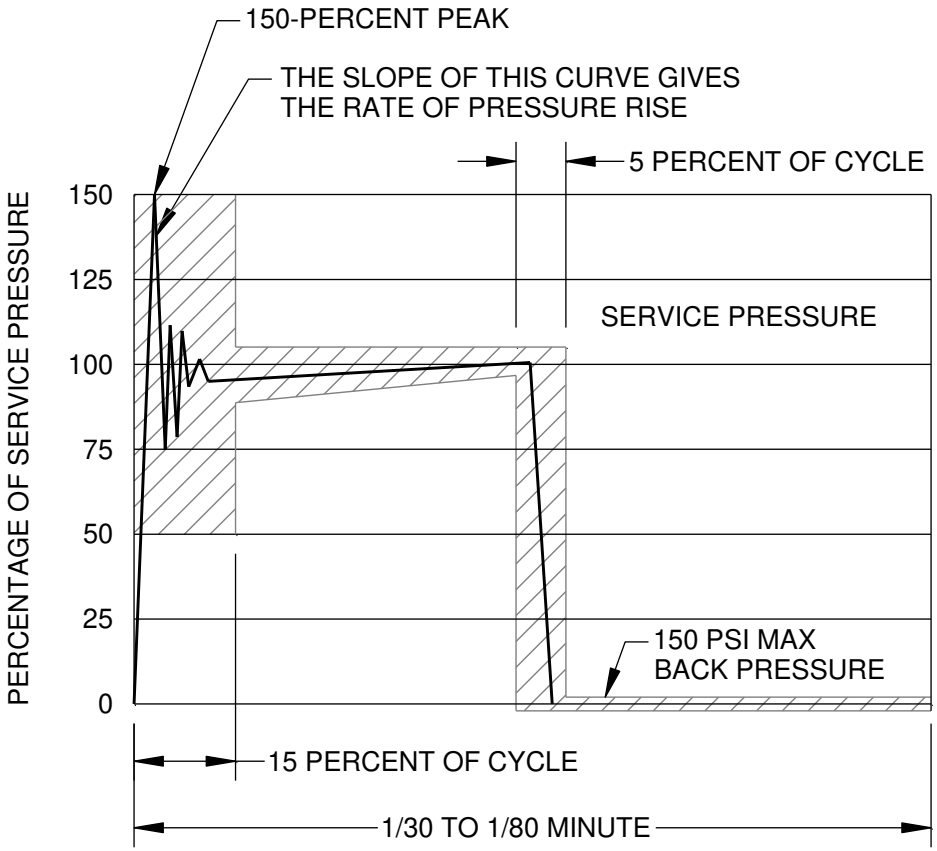
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NOTE: This curve is the approximate pressure time cycle for the pressure impulse to be used in performing the impulse test specified in 4.5.3. The actual pressure time cycle obtained when performing this test should fall within the shaded area.

FIGURE 2. Pressure time curve for impulse test (125 percent square wave).

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NOTE: This curve is the approximate pressure time cycle for the pressure impulse to be used in performing the impulse test specified in 4.5.3. The actual pressure time cycle obtained when performing this test should fall within the shaded area.

FIGURE 3. Pressure time curve for impulse test (150 percent peak wave).

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CONCLUDING MATERIAL

Preparing activity:
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